

# ChanYeong Hwang

[chanyeonghwang@gmail.com](mailto:chanyeonghwang@gmail.com) | +82 10 7518 4370 | [LinkedIn](#) | <https://CY-H1329.github.io>

## RESEARCH INTERESTS

---

My research focuses on advancing **Spatial Reasoning** as a core capability for **Physical AI**, bridging perception of an environment, real or virtual, with reasoning about how to act within it. I am drawn to **generalist agents** that adapt their reasoning across diverse goals and environments, a natural extension of the multi-agent orchestration work I have pursued. My goal is to advance the reasoning layer that lets agents plan grounded, generalizable actions across both physical and virtual worlds.

## EDUCATION

---

|                                                                         |                                 |
|-------------------------------------------------------------------------|---------------------------------|
| <b>Korea University</b> , Seoul, Republic of Korea                      | Sep. 2024 – Aug. 2026(Expected) |
| • Master of Science: Computer Science and Engineering (AI Applications) |                                 |
| <b>Sorbonne University</b> , Paris, France                              | Sep. 2022 – Jan. 2023(Dropout)  |
| • Master's Degree: Android/Artificial Intelligence                      |                                 |
| <b>Sorbonne University</b> , Paris, France                              | Sep. 2019 – May. 2022           |
| • B.S. in Computer Science and Mathematics                              |                                 |
| <b>Lycée Saint-Exupéry</b> , Ouagadougou, Burkina Faso                  | Sep. 2016 – Jul. 2019           |
| • High School Diploma: Baccalauréat With Honors                         |                                 |

## PUBLICATIONS

---

|                                                                                                                                     |           |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>SpatIO: Adaptive Test-Time Orchestration of Vision-Language Agents for Spatial Reasoning</b> <a href="#">[arXiv]</a>             | ECCV 2026 |
| <i>Chan Yeong Hwang</i> , Miso Choi, Sunghyun On, Jinkyu Kim, Jungbeom Lee                                                          |           |
| <b>Implementation of Two-Signature Security on an Identity Storage Smart Contract</b> <a href="#">[HAL]</a>                         | AINA 2025 |
| Daniel Maldonado-Ruiz, <i>Chan Yeong Hwang</i> , Elina Jankovskaja, Karima Sadykova, Messilva Mazari, Jenny Torres, Nour El Madhoun |           |

## RESEARCH EXPERIENCE

---

|                                                                                                                                                                                |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Graduate Researcher</b> , Vision & AI Lab – Korea University, Republic of Korea                                                                                             | Sep. 2024 – Current   |
| • Investigated <b>Machine Unlearning</b> methodologies to optimize model update efficiency and enhance data privacy.                                                           |                       |
| • Conducting research on <b>Spatial Reasoning</b> for Vision-Language Models (VLMs), focusing on developing frameworks for Mental Rotation Tasks (MRT) and 3D view generation. |                       |
| <b>Research Intern</b> , AI Applied Technology Team – VAIV Company, Republic of Korea                                                                                          | Mar. 2025 – Aug. 2025 |
| • Fine-tuned <b>DocLayout-YOLO</b> to develop a high-speed Korean Document Layout Analysis (DLA) system for RAG optimization.                                                  |                       |
| • Engineered a custom <b>1,000-page annotated dataset</b> to improve parsing accuracy for complex Korean document structures.                                                  |                       |

## WORK EXPERIENCE

---

|                                                                                                                                                      |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Engineer and Developer</b> , Autonomous A2Z – Anyang, Republic of Korea                                                                           | Apr. 2023 – Mar. 2025 |
| • Developed real-time video streaming and CAN communication pipelines for <b>remote driving systems</b> deployed across various vehicle platforms.   |                       |
| • Implemented low-latency UIs, enabling stable <b>intercontinental (<math>\approx 8,000</math> km)</b> teleoperation for autonomous vehicle testing. |                       |
| • Embedded in the <b>AI Research Team</b> for two months, supporting pattern recognition and data annotation for perception model training.          |                       |

## PROJECTS

---

**Decentralized Identity Storage Smart Contract**, Graduation Project Jan. 2022 – May 2022

Team Leader – Undergraduate graduation project at **Sorbonne University** integrated into the doctoral thesis  
"Implementation of Two-Signature Security on an Identity Storage Smart Contract" accepted at **AINA 2025** conference.

- Led a team of four to develop a decentralized identity management system on the **Ethereum** blockchain using **Solidity**.
- Engineered a **two-signature security protocol** (user and network signatures) to automate identity validation, effectively removing the need for a central administrator.
- Integrated **RSA asymmetric cryptography** for secure key generation and **SHA-256** hashing for cross-verifying data between the client-side (JavaScript) and the blockchain (Solidity).

**Finite Automata Programming with GUI** Jan. 2021 – Mar. 2021

Solo Project

- Developed a comprehensive finite automata simulator in **Python**, implementing core algorithms for state transitions, string acceptance, and automata minimization.
- Designed and built an intuitive **Graphical User Interface (GUI)** that allows users to visually model, edit, and simulate complex state machines in real-time.

**Java Table Tennis Game** Oct. 2020 – Nov. 2020

Team Member

- Created a physics-driven table tennis game in **Java**, featuring realistic collision detection and dynamic paddle-to-ball momentum transfer.
- Implemented **probability-based ball trajectory algorithms** to simulate varied spin and bounce patterns, significantly enhancing the gameplay experience.

**World School's Project: Burkina Faso Irrigation System** Sep. 2017 – Jul. 2019

Team Leader

- Spearheaded an international research collaboration with a partner high school in Germany to design sustainable field irrigation solutions for agricultural sites in **Burkina Faso**.
- Coordinated cross-border communications and project milestones, conducting technical feasibility studies on irrigation efficiency and localized resource management.

## EXTRA-CURRICULAR ACTIVITIES

---

**Teaching Assistant, Python & R Workshop – Ajou University, South Korea** Apr. 2023 – May. 2023

- Instructed non-CS majors on Python and R fundamentals, focusing on web data crawling, processing, and statistical analysis.
- Provided hands-on mentoring and led students through technical challenges, fostering academic leadership and a sense of professional pride.

**Volunteer, Dail Community (Babfor) – Seoul, South Korea** Oct. 2022 – Nov. 2022

- Served the elderly and homeless populations by assisting in kitchen operations, meal distribution, and facility maintenance.
- Developed values of service and empathy through collaboration with diverse volunteer groups and community engagement.

**Club Member, Robotech (Robotics Club) – Sorbonne University, Paris, France** Dec. 2019 – Sep. 2021

- Participated in a cross-disciplinary engineering community dedicated to robotics research and interdisciplinary technical discussions.
- Collaborated with peers from various academic backgrounds to solve technical problems and initiate team-based engineering projects.

**Part-time Server, Daso (Korean Restaurant) – Paris, France** Jan. 2019 – Dec. 2019

- Developed professional interpersonal relationships and social adaptability by engaging with a diverse international clientele, while mastering efficient service coordination in a fast-paced restaurant environment in Paris.

## LANGUAGES

---

**Korean:** Native

**French:** Native (DEL F/DALF C2)

**English:** Fluent (TOEFL/TOEIC)

**Spanish:** Intermediate

## SKILLS

---

**Programming:** Python, C/C++, Java, JavaScript / TypeScript, OCaml, Solidity

**AI & Data:** PyTorch, TensorFlow, Gym, NumPy, Pandas, Matplotlib

**Web Development:** React, Next.js, Electron, Zustand

**Design & Graphics:** Adobe Photoshop, GIMP

**Security & Blockchain:** RSA Cryptography, SHA-256 Hashing, Smart Contract Development